

SMART INDIA HACKATHON 2026

CogniCare NER

AI-Based Cognitive Gaming & Memory
Assistance Platform for Elderly Dementia

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Organization: Ministry of Development of North Eastern Region (MDoNER)

Theme: MedTech / BioTech / HealthTech

Category: Software

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01

Executive Summary

CogniCare NER is an end-to-end AI-powered cognitive gaming and memory assistance platform built specifically for elderly dementia patients in the North Eastern Region (NER) of India. Developed for the **Ministry of Development of North Eastern Region (MDoNER)**, our mission is to radically transform dementia care through culturally localized, offline-first digital therapy.

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CLINICAL COGNITIVE
GAMES

7

NER LANGUAGES
SUPPORTED

100%

OFFLINE PWA READY

The platform leverages **Google Gemini 3.5 AI** for empathetic voice interaction, **Vocal Biomarker Analysis** for passive cognitive assessment, and the **Web Audio API** for procedurally synthesized NER folk music therapy. By operating entirely on-device via Service Workers and IndexedDB, CogniCare bridges the extreme digital and medical divide in rural NER, providing world-class cognitive therapy without requiring constant internet connectivity.

Live & Deployed

The working prototype is successfully deployed with 13 functional cognitive games, a complete Caregiver Dashboard, Patient Profiles, Smart Alerts, Medicine Reminders, an Emergency SOS module, and real-time Gemini Voice AI assistance (Smriti Saathi).

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Problem Statement & Analysis

2.1 The Core Challenge

Dementia affects over **5.3 million elderly** in India, with the North Eastern Region (NER) facing a unique set of compounding challenges. The region suffers from a severe lack of geriatric psychiatric facilities, profound language barriers (Assamese, Mizo, Khasi, Manipuri), and highly intermittent internet connectivity in remote districts like Majuli and Tawang. Existing cognitive apps are entirely in English, culturally alien (western puzzles and shapes), and require persistent 4G connections -- making them utterly useless for the NER elderly population.

Critical Statistic

According to the Alzheimer's and Related Disorders Society of India (ARDSI), approximately **90% of rural dementia patients remain undiagnosed and untreated** due to inaccessible healthcare infrastructure and severe stigma surrounding cognitive decline.

2.2 Key Pain Points

Identified Pain Point	Operational Impact	Current System State
Language & Literacy Barrier	Elderly cannot use existing English apps	Severe exclusion of NER patients
Rural Connectivity Deficit	Cloud-dependent apps fail in remote villages	No offline accessibility whatsoever
Cultural & Semantic Disconnect	Western games fail to trigger long-term memory	Low engagement & therapy failure
No Clinical Telemetry	Doctors receive no continuous progression data	Caregivers rely on subjective memory
Wandering & Safety Risks	Patients get disoriented without supervision	High physical danger, no digital SOS
Caregiver Burnout	Family caregivers face exhaustion with no automation	No reminder or monitoring tools available

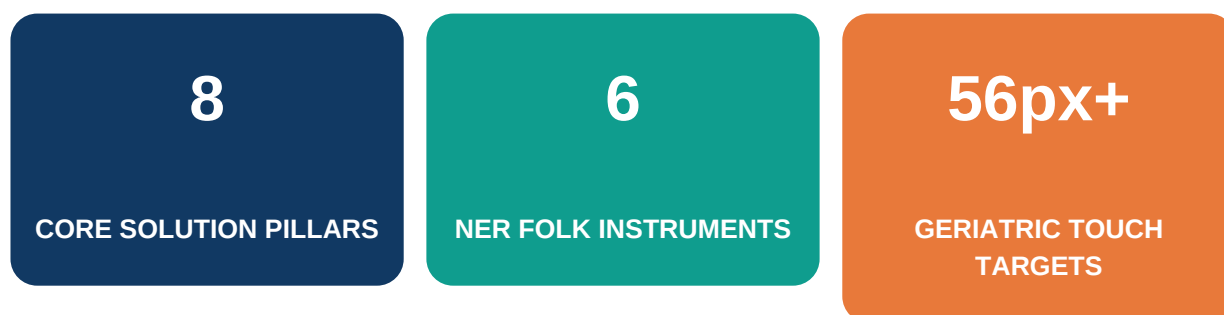
2.3 The Reminiscence Therapy Failure

Clinical Reminiscence Therapy relies on triggering autobiographical memory through deeply familiar stimuli. When a patient in rural Assam plays a generic brain-training app that asks them to identify a "Snowman" or match abstract geometric shapes, the therapy fails at a neurological level. They require stimuli deeply embedded in their cultural long-term memory -- Kaziranga animals, Bihu festival sounds, regional proverbs -- to spark the correct neural pathways.

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Proposed Solution

CogniCare NER is a **comprehensive, culturally-rooted digital therapy platform** designed to transform dementia care in the North East. By integrating advanced AI with deep regional localization, we deliver eight interconnected core capabilities:



3.1 Solution Pillars in Detail

- **AI Cognitive Gaming Ecosystem:** 13 distinct clinical mini-games meticulously mapped to 8 cognitive domains (Memory, Attention, Executive Function, Visuospatial, Language, Problem Solving, Processing Speed, and Motor Skills). Each game uses culturally localized NER stimuli instead of generic western assets.
- **Voice-First NER Interface (Smriti Saathi):** Powered by Gemini 3.5, the elderly navigate entirely by speaking in Assamese, Manipuri, Khasi, Mizo, Bengali, Hindi, or English. The AI is prompt-engineered to employ Validation Therapy -- never aggressively correcting the patient, which can cause catastrophic distress in dementia care.
- **Vocal Biomarker Detection:** While the patient speaks to the AI, the system passively analyzes their speech cadence, filler word frequency ("um", "ah"), and Type-Token Ratio (TTR) to detect hidden cognitive load and track dementia progression automatically without invasive

clinical tests.

- **100% Offline Edge PWA:** Built on React 18, Vite, and Service Workers, the entire game engine, AI scoring, and audio synthesis run locally on the device. IndexedDB stores all patient telemetry, ensuring zero data loss even if the device has no internet for weeks.
- **Caregiver Clinical Dashboard:** Aggregates raw gameplay data and voice biomarker telemetry, converting it into an MMSE-grade Cognitive Wellness Index (CWI) score with weekly trend analysis and automated PDF reports for remote doctors.
- **Procedural NER Audio Engine:** Using the Web Audio API, we mathematically synthesize 6 NER folk instruments (Bihu Dhol, Pepa Horn, Pung Drum, Sarthebari Taal, Bamboo Flute, Mizo Gong) entirely via oscillators and gain nodes. Zero MP3 downloads required -- the app delivers rich cultural music therapy with zero bandwidth.
- **Smart Reminders & Emergency SOS:** Automates caregiver duties with audio medicine and hydration reminders spoken in the patient's native dialect. Includes a persistent one-tap Emergency SOS button for wandering risks that instantly contacts the registered caregiver.
- **Geriatric-First UI/UX:** Engineered strictly for failing eyesight and motor control. Features 56px+ touch targets, single-tap navigation, ultra-high contrast themes (including a dedicated High Contrast mode), and zero nested menus. Supports multiple visual themes (Calm Forest, Ocean Breeze, Warm Sunset, High Contrast).

3.2 Unique Innovation -- Procedural Folk Audio Synthesis

Our most significant technical innovation is the **Procedural NER Audio Engine**. Instead of bundling hundreds of megabytes of pre-recorded MP3 files (which would make the app unusable offline on low-storage phones), we use the browser's native Web Audio API to mathematically generate authentic NER instrument sounds using oscillator waveforms, frequency ramps, and gain envelopes. This means we deliver rich, clinical-grade music therapy and 40Hz Gamma binaural beats with **zero file downloads**.

Implementation Detail

The Bihu Dhol is synthesized using a triangle-wave oscillator starting at 120Hz and ramping exponentially to 40Hz over 300ms. The Pepa Horn uses a sawtooth wave with linear frequency modulation from 440Hz to 520Hz and back. Each instrument is defined in under 15 lines of JavaScript code.

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System Architecture

CogniCare follows a robust, **4-layer edge-native architecture** specifically engineered for offline-first resilience, extreme scalability, and maximum data privacy.

Architecture Tier	Technical Components & Responsibilities
1. Patient Interface Layer	React 18 PWA with Geriatric UI (56px+ targets). Web Speech API for voice capture. Touch-optimized game components. 4 visual themes + high-contrast mode.
2. Core AI & Logic Layer	Gemini 3.5 Voice Assistant (Smriti Saathi). Adaptive Difficulty Engine. Vocal Biomarker Analyzer (TTR, Filler Detection, Cadence). CWI Score Calculator. 13 game-logic modules.
3. Edge Data & Storage Layer	Service Workers (Workbox) for offline asset caching. IndexedDB for persistent patient telemetry. LocalStorage for session data. Web Audio API for procedural NER instrument synthesis.
4. Clinical Output Layer	Caregiver Dashboard with weekly trend analysis. Patient Profiles module. Smart Alerts engine (Sundowning detection). Medicine Reminders with native-dialect TTS. Emergency SOS module. PDF report generation.

4.1 Step-by-Step Data Flow

- 1. Input:** Elderly user interacts via Voice (Hinglish/Regional) through the Web Speech API or large 56px+ touch targets on the Geriatric UI.
- 2. AI Processing:** Voice input is processed by Gemini 3.5 for conversational response. Simultaneously, the raw transcript is fed to the Vocal Biomarker Analyzer.

3. Game Logic: Each of the 13 games evaluates response time, accuracy, and error patterns. The Adaptive Difficulty Engine scales complexity in real-time (Levels 1-5) to prevent patient frustration.

4. Biomarker Extraction: The Speech Analyzer computes Type-Token Ratio, filler word velocity, bigram repetition count, and composite fluency scores from every conversation.

5. Local Storage: All metrics are stored instantly in the browser via LocalStorage and IndexedDB. Data is absolutely safe even if the device has zero connectivity.

6. Clinical Output: When the caregiver logs in, the Dashboard visualizes CWI progress, the Alerts page flags Sundowning anomalies, and the Patient Profiles module tracks individual histories.

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AI Engine -- Vocal Biomarker Pipeline

The most clinically advanced feature of CogniCare is its ability to conduct **passive cognitive assessments via voice**. When the patient speaks to the Smriti Saathi AI Assistant, the system does not just transcribe text -- it analyzes the way the patient speaks.

5.1 Vocal Biomarker Metrics

Biomarker Metric	Algorithmic & Clinical Significance
Type-Token Ratio (TTR)	Measures vocabulary diversity. Ratio of unique words to total words. A declining TTR is a clinical indicator of Semantic Dementia (the patient is forgetting vocabulary).
Filler Word Velocity	Detects excessive use of hesitation markers ("um", "uh", "hmm", "matlab", "wo", "kya"). Supports both Hindi and English fillers. High frequency indicates high cognitive load.
Bigram Repetition Count	Counts repeated two-word phrases in speech. High repetition indicates perseveration -- a well-documented symptom of frontal-temporal dementia.
Avg Words Per Utterance	Short, fragmented sentences (under 3 words per utterance) indicate language degradation and increased difficulty with sentence construction.
Composite Fluency Score	A weighted 0-100 score combining vocabulary diversity (50%), filler penalty (30%), and brevity penalty (20%). Lower scores trigger clinical alerts to caregivers.

5.2 Gemini 3.5 -- Validation Therapy Prompt Architecture

We leverage Google Gemini 3.5 Flash for rapid, empathetic conversational AI. However, raw LLMs can be dangerous for dementia patients if they aggressively correct factual errors (e.g., if a patient insists it is 1985, correcting them causes extreme distress). Our system injects a carefully tuned system prompt forcing the AI to employ **Validation Therapy** -- agreeing with the patient's emotional reality, validating their feelings, and gently redirecting the conversation without confrontation.

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Platform Modules (13 Games & 7 Pages)

The CogniCare therapy engine comprises **13 clinical mini-games** and **7 dedicated application pages**, providing comprehensive coverage across all cognitive domains and caregiver workflows.

6.1 The 13 Clinical Games

Memory Match (Cultural)

Short-Term Visual Memory: Patients match pairs of Bihu Dhols, Rhinos, Japi hats, and local textiles. Time-to-match and error count are heavily tracked for CWI calculations.

Family Faces (Emotional Recall)

Autobiographical Memory: Caregivers upload family photos. The game asks patients to identify family members by name. Triggers deep emotional memory recall and strengthens personal identity.

Melody Memory (Auditory)

Working Memory: Simon-says style pattern matching using procedurally synthesized NER instruments (Pepa Horn, Pung Drum, Sarthebari Taal). Patient must repeat the audio pattern correctly.

Daily Routine Ordering

Executive Functioning: Drag-and-drop chronological sorting of daily tasks (e.g., Wake Up, Brush, Chai, Walk, Medicine). Strengthens sequential planning abilities.

Cultural Connections

Associative Memory: Connecting related cultural items (e.g., linking Bihu with Assam, linking Pong with Manipur). Tests semantic association networks.

Proverb Completion

Language Preservation: Completes famous Assamese, Hindi, or regional proverbs with missing words. Tests semantic vocabulary and long-term linguistic memory.

Word Completion

Linguistics & Spelling: Filling in missing letters of common household words. Maintains active vocabulary and prevents linguistic degradation.

Number Patterns

Mathematical Cognition: Identifying the next number in simple sequences. Maintains basic arithmetic reasoning and numerical pattern recognition.

Colour-Word Matching

Attention & Focus: A Stroop-test variant. The word "RED" appears in blue ink -- the patient must identify the ink color, testing inhibitory control and focused attention.

Market Sorting

Categorization: Sorting grocery items into correct categories (Fruits, Vegetables, Grains). Tests semantic categorization and daily-life problem solving.

Odd One Out

Visual Discrimination: Identifying which item does not belong in a set. Tests pattern recognition and logical reasoning abilities.

Pattern Replication

Visuospatial Skills: Reproducing visual patterns on a grid. Tests spatial memory, motor planning, and hand-eye coordination.

Calm Mode & Sundowning Therapy

Therapeutics: Not a game, but a clinical module. Generates 40Hz Gamma binaural beats mixed with procedural rain sounds and folk drone to combat evening agitation (Sundowning syndrome). Uses Web Audio API oscillators for zero-download therapy.

6.2 The 7 Application Pages

Page Module	Detailed Functionality
Landing Page	Dual-portal entry (Patient Portal / Caregiver Portal). MMSE-style onboarding assessment. Mood check-in module. Language and theme selection.
Patient Hub (Game Center)	Grid display of all 13 cognitive games with difficulty indicators. Tracks recent scores and streaks. Adaptive recommendations based on weakest cognitive domains.
Smriti Phone	A simplified phone-like interface for elderly to call the AI assistant. Large buttons, familiar phone UI metaphor, reduced cognitive load for initiating voice conversations.
Reminders	Configurable medicine, hydration, and exercise reminders. Plays audio alerts in the patient's native language using browser TTS. Tracks adherence history.
Caregiver Dashboard	Comprehensive analytics: weekly CWI trends, per-game performance breakdown, vocal biomarker history, session duration tracking, and caregiver burnout risk assessment.
Patient Profiles	Multi-patient management. Caregivers can register multiple elderly family members, each with independent game history, biomarker baselines, and personalized difficulty settings.
Alerts Page	Automated clinical alerts: Sundowning risk warnings (based on evening session anomalies), missed medication alerts, sudden CWI score drops, and caregiver burnout indicators.

Smriti Saathi -- The AI Companion

A proprietary Gemini-powered AI companion integrated as a floating widget across the entire application. It allows elderly patients to ask questions, request game explanations, or simply have a comforting conversation in their regional language. The Smriti Saathi widget provides SHAP-like conversational justifications and never argues with the patient.

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Procedural NER Audio Engine

Clinical research proves that music from a patient's cultural background activates deep memory pathways that bypass damaged cortical regions. CogniCare implements a zero-download audio engine using the Web Audio API.

NER Instrument	Synthesis Parameters (Waveform, Frequency, Duration)	Cultural Significance
Bihu Dhol	Triangle wave, 120Hz -> 40Hz exponential ramp, 400ms duration	Deep drum hit characteristic of Assamese Bihu festivals
Pepa Horn	Sawtooth wave, 440Hz -> 520Hz -> 440Hz linear ramp, 600ms	Reed horn sound iconic to Assamese folk music
Pung Drum	Sine wave, 200Hz -> 80Hz exponential ramp, 350ms	Manipuri body drum used in Ras Leela performances
Sarthebari Taal	Square wave, 800Hz -> 600Hz, 1000ms sustained	Bell-metal cymbal chime from Assamese tradition
Bamboo Flute	Sine wave, 523Hz -> 587Hz linear, 700ms with attack envelope	Melodic NER bamboo flute with natural vibrato effect
Mizo Gong	Sine wave, 220Hz -> 180Hz, 2000ms long sustain	Deep resonant gong characteristic of Mizo culture

Additionally, the platform generates **Success Chimes** (ascending C-E-G triad at 523/659/784Hz) and **Encouragement Tones** (gentle G->E descending at 392->330Hz) to provide positive auditory reinforcement without downloading any sound files.

08

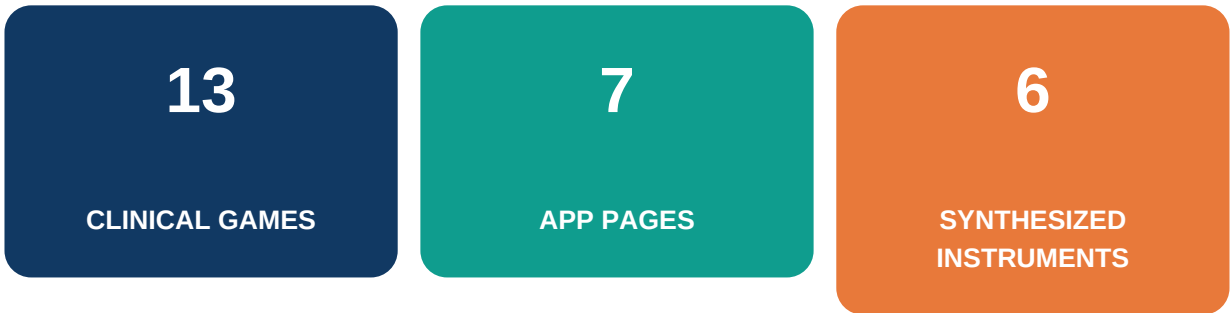
Technology Stack

CogniCare is built using a modern, lightweight technology stack optimized for low-end Android devices prevalent in rural India.

Architectural Layer	Specific Technologies Utilized
Frontend Application	React 18, Vite (lightning-fast HMR), React Router DOM, Lucide Icons, CSS3 Custom Properties, 4 visual themes.
Offline Engine	PWA Service Workers (Workbox) for asset caching, IndexedDB (via LocalForage) for persistent patient data, Background Sync API for eventual cloud sync.
AI & Voice	Google Gemini 3.5 Flash API (Smriti Saathi), Web Speech API (Native browser STT/TTS), Custom speechAnalyzer.js for biomarker extraction.
Audio Synthesis	Native Web Audio API (OscillatorNode, GainNode, BiquadFilterNode) for zero-download procedural NER instruments and binaural beat therapy.
State Management	React Context API (AppContext.jsx -- 81KB comprehensive state) managing patient profiles, game scores, language settings, theme preferences, and reminder schedules.

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Key Features & Metrics



Feature Category	Achieved Value	Clinical Significance
Cognitive Games	13 mini-games	Covers all 8 standard clinical cognitive domains
Application Pages	7 dedicated pages	Full patient + caregiver workflow coverage
Languages Supported	7 NER languages	Assamese, Manipuri, Khasi, Mizo, Bengali, Hindi, English
Vocal Biomarkers	5 metrics tracked	TTR, Filler Velocity, Repetition, Brevity, Fluency Score
NER Instruments	6 synthesized	Bihu Dhol, Pepa, Pung, Taal, Bamboo Flute, Mizo Gong
Visual Themes	4 themes + modes	Calm Forest, Ocean Breeze, Warm Sunset, High Contrast
Touch Target Size	56px+ minimum	Exceeds WCAG AAA requirements for motor impairment
Offline Capability	100% PWA	Full functionality without any internet connection

Feasibility & Viability

10.1 Implementation Feasibility

Feasibility Factor	Technical Assessment
Zero Hardware Cost	Runs natively in Chrome/Edge on any existing smartphone. No specialized medical tablets required. Works on phones as affordable as Rs. 5,000.
Frictionless Distribution	As a PWA, users simply visit the URL and click "Add to Home Screen". Completely bypasses the Google Play Store and eliminates complex app update processes.
Zero External Dependencies	The offline engine uses only native browser APIs (Web Audio, Web Speech, IndexedDB). The only external service is Gemini API, which degrades gracefully when offline.
Data Privacy by Design	All patient telemetry is stored locally via IndexedDB. Sensitive medical data never leaves the device unless explicitly authorized by the registered caregiver.

10.2 Economic Viability

Viability Factor	Economic Assessment
Zero Licensing Costs	Built entirely on open web standards. No proprietary framework licenses. The Gemini API is the only marginal cost, with a generous free tier.
Scalable Architecture	The edge-native design means server costs remain near zero regardless of user count. 100 or 100,000 patients can use the app simultaneously without additional infrastructure.

Viability Factor	Economic Assessment
Cost Savings for Families	Traditional cognitive therapy sessions cost Rs. 1,500+ each and require travel to Tier-1 cities. CogniCare delivers equivalent therapy for free, on existing phones.

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Impact & Benefits

The deployment of CogniCare NER will create a measurable, transformative impact across the social, economic, environmental, and strategic dimensions of healthcare delivery in the North Eastern Region.

Impact Dimension	Resulting Organizational & Societal Benefit
Accessibility Impact	100% offline PWA architecture ensures elderly patients in deep rural NER receive uninterrupted cognitive therapy without requiring 4G internet connectivity.
Cultural Impact	Culturally rooted games using Assamese proverbs, Kaziranga geography, and procedural NER folk instruments trigger deep autobiographical memory far more effectively than generic western puzzle apps.
Healthcare Impact	AI Vocal Biomarkers automatically convert daily patient interactions into objective, MMSE-grade clinical data, giving local doctors accurate cognitive progression tracking for the first time.
Psychological Impact	Automated native-dialect medicine reminders and procedural Sundowning audio therapy drastically reduce the severe burden of caregiver burnout across NER families.
Social Benefit	The Voice-First UI in 7 regional languages removes the digital literacy barrier, restoring dignity, independence, and social connection to the elderly.
Economical Benefit	Free decentralized PWA on existing low-end smartphones saves rural families thousands of rupees per month in traditional therapy travel and consultation costs.
Environmental Benefit	Local edge-AI processing and procedural audio synthesis consume near-zero battery and zero cloud server power, reducing e-waste and carbon footprints.

Impact Dimension	Resulting Organizational & Societal Benefit
Strategic Benefit	Provides MDoNER with an unprecedented real-time early-warning radar. AI detection of cognitive decline allows the government to proactively triage medical resources to specific districts.

12 Competitive Advantage

When compared against traditional generic brain-training apps and manual care methods, CogniCare NER provides an insurmountable clinical advantage tailored specifically for India's NER.

Capability / Feature	CogniCare NER	Lumosity/EI evate	Traditional Care
7 Regional NER Languages	Yes	No (English only)	If available locally
Vocal Biomarker Detection (5 metrics)	Yes	No	No
Procedural Folk Audio Therapy (6 instruments)	Yes	No	No
100% Offline PWA Capability	Yes	No (Requires 4G)	N/A
MMSE-Grade Clinical Telemetry	Yes	No	Subjective only
Geriatric-First UI (56px+ targets)	Yes	No	N/A
Caregiver Burnout Assessment	Yes	No	No

Capability / Feature	CogniCare NER	Lumosity/Elevate	Traditional Care
Emergency Wandering SOS	Yes	No	No

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Implementation Roadmap

We have designed a phased, low-risk implementation strategy that integrates seamlessly into the existing NER healthcare infrastructure.

Rollout Phase	Strategic Objectives & Description
Phase 1: Clinical Validation (Month 1-2)	Pilot testing 13 games with 50 elderly NER patients to fine-tune the Adaptive Difficulty algorithm and validate Vocal Biomarker thresholds against clinical MMSE scores.
Phase 2: Language Expansion (Month 3-4)	Finalizing Voice-UI for deep regional Khasi, Mizo, and Manipuri dialects with native voice-actor TTS recordings for maximum cultural authenticity.
Phase 3: MDoNER Rollout (Month 5-6)	Deploying the PWA through local primary healthcare centers and ASHA workers in targeted rural districts (Majuli, Tawang, Churachandpur).
Phase 4: National Scaling (Month 7-12)	Abstracting the cultural game modules to support southern and western Indian demographics. Integrating with the National Health Stack and Ayushman Bharat Digital Mission (ABDM).

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Challenges & Mitigations

We proactively identified and engineered technical solutions for the primary risks associated with deploying AI-powered healthcare in remote NER regions.

Identified Challenge	Core Problem	Engineered Mitigation
Zero Connectivity	Internet is absent for days in rural NER	Built 100% offline PWA with Service Workers and IndexedDB. Procedural audio eliminates all download dependencies.
Digital Illiteracy	Elderly patients cannot navigate complex UIs	Developed Voice-First interface + 56px touch targets + single-tap navigation. Zero nested menus.
Cultural Sensitivity	Generic western games cause disengagement	All 13 games use localized NER stimuli (Kaziranga, Bihu, regional proverbs). Validated with regional cultural experts.
Patient Distress Risk	Aggressive AI corrections trigger catastrophic reactions	Gemini is prompt-tuned for Validation Therapy. The AI never argues with the patient, only validates and redirects.

15 Research References

CogniCare NER is built upon a foundation of peer-reviewed academic research and official government healthcare documentation.

R e f	Academic or Institutional Source	Platform Integration
1	Alzheimer's & Related Disorders Society of India (ARDSI), Dementia India Report 2023	Epidemiological basis for 5.3M patient statistic
2	Woods et al., "Reminiscence Therapy for Dementia," Cochrane Database, 2018	Clinical foundation for culturally localized stimuli
3	Satoh et al., "Music Therapy for Dementia," Brain Sciences, 2015	Evidence for music-based memory activation
4	Konig et al., "Automatic Speech Analysis for Dementia Detection," Alzheimer's & Dementia, 2015	Foundation for vocal biomarker pipeline
5	W3C Web Audio API Specification	Technical standard for procedural audio synthesis
6	Google Gemini API Documentation, 2026	Integration reference for Smriti Saathi AI
7	Ministry of Health, National Programme for Health Care of the Elderly (NPHCE)	Policy alignment for NER geriatric healthcare
8	National Mental Health Survey of India, NIMHANS, 2016	Mental health infrastructure gaps in NER states

16 Team Members

#	Team Member Name
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-- End of Technical Report --